Cover Shadow Value: measuring the value of cover shadows with tracking and body orientation

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ABSTRACT

A winger who makes almost no tackles or interceptions can still be decisive in how a team defends, because his positioning and body orientation close passing lanes and steer the opponent towards where the team wants to press. This contribution, so far perceptible mainly through observation, does not show up in traditional defensive metrics, which only record what happens with the ball. This work proposes and demonstrates a way to measure it, the Cover Shadow Value, which uses SkillCorner's open tracking and body pose data (two A-League 2024/25 matches) to identify, with a cone defined by trunk orientation, the passing lanes each player closes with his body and the value those lanes would have, adding the danger the pass would create to the risk of losing the ball when attempting it. In an independent review, carried out by a coach external to the work on new plays, the shadows were considered coherent in 8 of 8 plays and the marking correct in 7 of 8. The trunk-based shadow, in turn, helps predict where the ball carrier passes and, in a model with both shadows, keeps the weight the ball-facing shadow loses, although its out-of-sample advantage over the latter is small, while the data do not allow any conclusion about pass success. With only two matches with pose and an arrival index that could not be calibrated as a probability, the results are not meant to evaluate players, but they show that this action can be measured, leaving validation to those with more data.

Keywords: cover shadow, body orientation, tracking, body pose, expected threat, Shapley value.

1. Introduction

1.1 Context and problem

When talking about the defensive moment, or the out-of-possession moment, one usually thinks of ball recoveries, interceptions or duels. It is also worth considering, however, how the opponent is conditioned, not least because some players with a considerable defensive impact hardly ever win the ball back, since their positioning and body orientation, when correct, close passing lanes and steer the opponent towards a given channel.

An example helps make this concrete. Consider a team out of possession, organised in a mid block, with the ball at an opposing centre-back. At that moment the winger has, broadly speaking, two options: stay wide, close to the opposing full-back, or position himself inside, closing the inside passing lanes and orienting his body so as to steer the ball wide, an area where the team can press more effectively. Decisions of this kind are central to a coach's work and often weigh on how well a team defends.

Traditional defensive metrics (tackles, interceptions and the like), however, do not capture the value of these actions, because they only record what happens with the ball. A winger with few tackles or interceptions may thus be seen as having a light defensive workload while being, at the same time, essential to the way the team defends.

Some work already tries to value this kind of action, such as Cascioli et al. (2025), with StatsBomb 360 data. In that work, however, the cover shadow is defined from the passing lane and the defender's reach, on static

snapshots taken at the moment of the pass, without considering body orientation, a component that the present work seeks to add, together with some refinements to the definition.

This requires defining with data what a cover shadow is, built from the defender's positioning and body orientation, distinguishing it from man-marking, assigning a value to a passing lane closed by that shadow and sharing that value out when more than one defender contributes to closing it. The scope of the project is therefore limited to demonstrating one possible approach to quantifying this action, whose body component has so far been perceptible mainly through observation. The central contribution is the cone defined by the defender's trunk orientation, supported by continuous measurement throughout possessions, the sharing of each lane's value among the defenders who close it, through the Shapley value, and a lane value that adds the cost of losing the ball to the danger of the pass.

1.2 Objectives and research questions

The general objective is to demonstrate how the value of off-ball defensive positioning can be measured with tracking and body pose data through cover shadows, with the cone defined by the defender's body orientation. Five specific objectives follow from it: to operationally define a cover shadow, to distinguish it from man-marking, to assign a value to a closed passing lane and share it among the defenders who close it, to evaluate the stability of the metric and the contribution of body orientation, and to communicate the method in a way coaches and analysts can follow.

The work seeks to answer five research questions, the last added in the second loop of the process (section 3.1):

1. How can a cover shadow be operationally defined from tracking and body pose data?
2. How can a value be assigned to a closed passing lane and shared among the defenders who close it?
3. Does body orientation change the shadow compared with a definition based only on the position relative to the ball?
4. Is the metric stable under reasonable variations of its parameters?
5. Does the trunk-based shadow help predict pass choice?

2. Related work

2.1 The cover shadow as a tactical concept

In popular tactical writing, the cover shadow is described as the area behind a defender that is blocked from the ball carrier's view, so that a pass to a player standing in it is no longer available (The Football Analyst, 2025). The same source links it to pressing the ball carrier, to closing the passing lane to a second opponent at the same time and to steering play towards a predictable area, functions that in practice are rarely separated. Coach education material describes the shadow in a similar way, as a triangle- or cone-shaped area behind the defender, corresponding to the space he covers and into which the attacker cannot make an unobstructed pass (Schamberg, 2022), the shape the metric proposed here adopts.

More relevant to this work is the fact that this literature explicitly links the shadow to body orientation. Curved pressing runs would create a larger shadow than running straight at the ball carrier and, from the same position, a defender turned slightly inwards would block the pass to the defensive midfielder, while a defender opened up to the outside would block the route to the full-back (The Football Analyst, 2025). Body orientation is therefore part of the tactical definition of the concept, an idea that has not yet reached quantitative models.

2.2 Valuing actions with expected threat

Expected threat (xT), proposed by Singh (2018), divides the pitch into a grid and assigns each zone the probability of the team scoring in the following actions, computed iteratively from how often players shoot from each zone or move the ball to other zones. The value of a pass or a carry is the difference between the xT of the destination zone and that of the origin zone, which makes it possible to measure what an action gains even when it does not end in a shot. The original version uses a 16 × 12 grid; this work uses the open 12 × 8 version, made available by the same author.

For measuring a shadow, xT has a convenient property: it depends only on the zone of the pitch and not on the defenders' positions, so the value of a passing lane does not change when a defender is removed to estimate the effect of his shadow. The trade-off is that it ignores the context of the play and carries values estimated in another league.

This work uses xT in two ways. The first is the usual one, the gain of a passing lane, which nonetheless has a limitation for the problem at hand, since zones close to one's own goal are worth almost nothing and a short pass from the goalkeeper to a centre-back therefore has a near-zero gain, even though closing it is one of the tasks most asked of players pressing high. The second reads the grid from the opponent's point of view and estimates the danger the opponent would gain by winning the ball back at the point where the lane would be cut, a cost that grows the closer to one's own goal the turnover happens (section 6).

2.3 Interception time and space control

Pitch control models estimate, for each point on the pitch, which player would reach the ball first. In the formulation of Spearman (2018), as implemented by Shaw in the open Friends of Tracking repository, each player takes a reaction time of 0.7 s, during which he keeps the velocity he had, and then runs in a straight line at 5 m/s; the ball travels at 15 m/s and the interception probability comes from a logistic function of the difference between arrival times, with an uncertainty of 0.45 s. This work adopts these values as a starting point, not least because they have a published origin and can be checked in the code.

2.4 The off-ball defensive contribution

The idea of crediting the defender who prevents an action, and not only the one who wins the ball back, has precedents. Kothari (2021) proposed a pass prevention measure that combines pitch control with possession value, crediting each defender with the difference in an attacker's control when that defender is removed from the calculation. Bischofberger et al. (2026), in turn, distribute changes in expected threat among individual defenders based on defensive pressure areas and on baselines conditioned on each player's role in the team structure, using data from 516 matches across three competitions.

More recently, Kim et al. (2025) decomposed the expected value of a possession into three components (action choice, its success and the value of the outcome) and assigned each defender a share of the changes in that value, in categories that include interception and deterrence, that is, credit for leading the opponent to choose a less dangerous action, using graph neural networks trained on tracking data from 564 Eredivisie matches. Of the known approaches, it is the closest to the idea of valuing how the ball carrier's decision is conditioned, although body orientation is not among the variables used, which are limited to positions, velocities and distances.

None of these approaches, however, isolates the cover shadow as an action: they measure a defender's control of, or responsibility for, space, not the closing of a specific passing lane by his body.

2.5 Body orientation in passing models

Arbués-Sangüesa et al. (2020) estimated players' orientation from video through pose estimation and used it to assess pass feasibility, combining how the ball carrier and possible receivers are oriented towards each other with the position of the opponents, which, over more than 6,000 passes, placed the pass actually made among the three most likely options in over 70% of cases. Orientation enters, however, on the attacking side: what is measured is whether the players of the team in possession are facing each other, not how the defender's body conditions the ball carrier's options.

2.6 Cover shadows measured with data

The closest work is Cascioli et al. (2025), which defines the cover shadow as the defender occupying the space between the ball carrier and a possible receiver, excluding those in man-marking (a defender within a 3 m radius centred 1 m behind the attacker, on the goal side). The authors compare simple geometric corridors with a physical model, which they call Lane Control, that compares the arrival times of the ball, the defenders and the receiver at points along the passing lane, and they credit each shadow with the difference in threat between the real scenario and the scenario without it, combining xT with pass selection and success probabilities estimated by neural networks.

The authors themselves point out the limits of the data used: static snapshots (StatsBomb 360) at the moment of each event, without measured velocities and with player identity in only one match, and they propose the use of continuous tracking as future work. The model does not consider the defender's body orientation, and the case in which several defenders close the same lane is not handled explicitly (removing one defender at a time, both tend to be undervalued, since the other keeps the lane closed).

2.7 The gap this work seeks to fill

The tactical literature links the cover shadow to body orientation, but the available quantitative models define it only by the defender's position and reach, and, in the works consulted, body orientation enters on the attacking side. SkillCorner's release of continuous tracking with player identity and three-dimensional body pose, albeit for only two matches, makes it possible to test a definition in which the shadow cone depends on the defender's trunk, measured continuously throughout possessions and with the value shared among defenders through the Shapley value (Shapley, 1953), and to check whether that orientation influences the ball carrier's decision. It does not follow that body orientation makes the metric more correct in every case, a question evaluated in section 7.

3. Methodology

3.1 Study design and adaptation of CRISP-DM

This is an applied study that proposes and demonstrates a method, not an evaluation of players or teams. The process followed CRISP-DM (Chapman et al., 2000), adapted to the problem: problem understanding corresponds to section 4, data understanding and preparation to section 5, modelling to section 6 and evaluation to section 7. The conclusions close the document but are not an additional phase.

Since there is no target variable (a blocked pass is never observed), the metric was neither trained nor validated with a train–test split. Validation rests instead on criteria set in advance: review of plays by the author and by external coaches, coherence with the passes actually made, stability under variations of the parameters, comparison with an alternative without body orientation, reproducibility of the results and, in the second loop of the process, the ability to predict pass choice. The auxiliary models introduced in that second loop (pass choice and the probability of the pass arriving) were, by contrast, fitted and validated out of sample, always leaving one match out at a time.

The process was iterative, as CRISP-DM itself recommends. The first evaluation led to six changes to the definition and the second to two more, giving rise to three versions of the metric (v1, v2 and v3), all recorded, with their justification, in a decision log that accompanies the code. A second loop, prompted by a coach's reading of one of the plays (section 8), added a rule for long passes, a pass choice model and a new way of valuing each lane, giving rise to the final version, v3.1 (the four versions are compared in Figure 1). A more ambitious version, which separated the value of cutting the lane from the value of making the ball carrier change his mind, was tested against criteria fixed before computation and was not adopted (section 7).

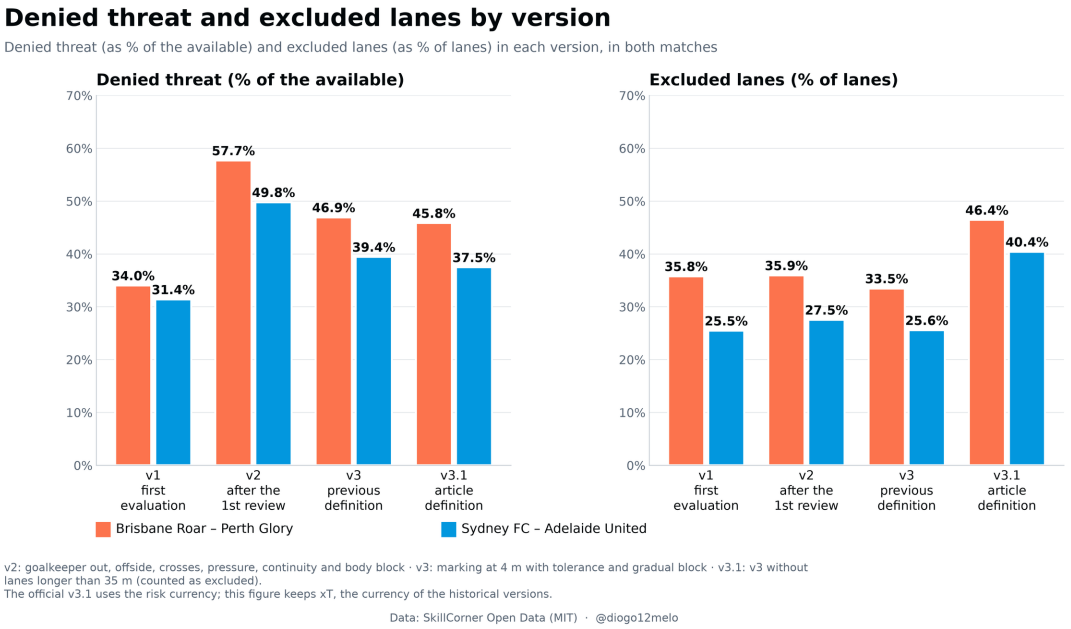


Figure 1. Denied threat and excluded lanes by metric version and match (v1, v2, v3 and v3.1), in xT, the way of valuing lanes used in the earlier versions.

3.2 Data

The work uses SkillCorner's open data (SkillCorner, 2025; MIT licence), which include 20 A-League 2024/25 matches with tracking at 10 frames per second, extracted from the television broadcast, dynamic events (possessions, passing options) and phases of play. Body pose, with 29 joints in three dimensions at 25 frames per second, exists for only two of those matches: Brisbane Roar–Perth Glory (21/12/2024) and Sydney FC–Adelaide United (01/02/2025). Since the definition of the metric rests on body orientation, the metric is computed only for these two matches, the remaining 18 being used only to fit and validate a model of the probability of a pass arriving, which does not depend on pose (sections 5.5 and 7).

Pose is only estimated for the players the camera is capturing, while tracking also estimates the position of the others, a limitation that constrains which passing lanes can be evaluated (section 5).

3.3 Reproducibility

The code was written in Python, organised in one notebook per CRISP-DM phase and in modules with automated tests, with all the metric's parameters gathered in a single file. Preparation and computation run from the original data without manual intervention, and both matches were computed with the same parameters, with no adjustments specific to either.

4. Problem understanding

4.1 Success criteria

With no target variable, the success of the metric was defined in advance by seven criteria, to which the second loop of the process added an eighth, all evaluated in section 7. The first seven were defined as qualitative questions, with no numerical threshold set in advance, so the verdicts in section 7.10 are the author's judgements, presented with the evidence that supports them, while the eighth, like the tests in sections 7.7 and 7.8, rests on a comparison fixed before computation.

Table 1. Success criteria defined before modelling.

Criterion	What is sought	How it is checked
K1	Face validity: a coach considers the shadows coherent	Review of plays: do the shadows and values make sense?
K2	Separates shadow from marking	In the same review: are the defenders classified as marking actually marking?
K3	Coherence with the game (sanity check, not a conclusion)	Lanes in shadow should be used less and fail more in real passes
K4	Stability	Reasonable variations of the parameters do not drastically change the results
K5	The body changes the shadow	Comparison with a cone facing the ball, without body orientation: in how many lanes does the defender casting the shadow change
K6	Reproducible and traceable	The computation can be repeated from the original data and every number traces back to the data
K7	Explainable	Coaches understand the animation of a play in under a minute
K8	The body influences the ball carrier's decision	A pass choice model predicts better, out of sample, with the trunk-based shadow than with no shadow and than with the ball-facing shadow

4.2 Constraints, risks and assumptions

The main constraints come from the data: pose in only two matches of one league, pose only for the players captured by the camera and no video. The risks identified at the outset were error in the joint estimates, misalignment between tracking and pose (generated separately by SkillCorner) and the existence of few moments with all relevant defenders having pose. The metric is assumed to refer to ground passes up to 35 m, the value of each lane to be approximated by the xT gain plus the danger the opponent would gain by winning the ball back at the cut point (section 6), and the cone length to represent the distance behind the defender up to which the ball cannot be played over him.

The data are public and were used with credit to the source. Since the aim is to demonstrate a method, player-level results are presented only as illustration, without judgements about specific players.

5. Data understanding and preparation

5.1 What the data have and lack

In a first 12-second sample, 50% of player–frame combinations had pose and, whenever it existed, it included all 29 joints, each with an error estimate. Pose only exists for the players detected by the camera, although tracking gives everyone's position, and over the whole match coverage is lower: considering only defenders during possessions, pose covers 64.5% of cases in the Brisbane–Perth match and 73.1% in the Sydney–Adelaide match. Coverage increases with proximity to the ball (in the first match, 82.8% of defenders within 30 m and 92.7% within 20 m), which is understandable, not least because the broadcast camera follows the ball.

The data include no video and pose was generated separately from tracking, at different frequencies (25 and 10 frames per second), so the alignment between them had to be checked before any computation.

5.2 Alignment between tracking and pose

The rule given by SkillCorner (pose frame = 2.5 × tracking frame) was confirmed: timestamps match and the distance between each player's two positions is minimal with no offset at all. Every five pose frames, the two sources coincide exactly, which gives five moments per second without interpolation, a frequency adopted for the whole computation. In possessions too short to contain one of those moments (130 of the 1,054 possessions in the first match, almost all one-touch plays), the nearest aligned moment, 0.1 s away, is used.

The check also showed that, when the camera does not capture a player well, the tracking position can drift away from the body reconstructed by pose: the median difference is 0.19 m for detected players but rises to 0.43 m for undetected ones, with cases above 12 m. For that reason, whenever pose exists the position that follows the body is used, and the tracking position only in its absence.

5.3 Body orientation

Three ways of measuring which way a defender is facing were compared: the perpendicular to the shoulder line, the same idea applied to shoulders and hips together (the trunk) and the direction of the head, from the ears to the nose. The first two were validated by the position of the nose, which lies in front of the neck in 99.9% of cases, and agree with each other (99.2% of differences below 20°).

The trunk proved to be the most stable version, with a median change of 6.6° between measurements 0.2 s apart, against 11.1° for the shoulders and 8.2° for the head. The head deviates from the trunk by about 18° (median), which corresponds to the defender scanning the game around him, and it was therefore left out of the cone definition, the trunk being kept as the reference for body orientation.

5.4 Moments and evaluated lanes

The working table gathers, per player and aligned moment, the position, detection, pose and the head, shoulder and hip joints, only during possessions: 8,915 moments (29.7 minutes of possession) in the first match and 8,145 in the second. A passing lane is only evaluated when all the defenders who can affect it have pose; the others are left out and counted, which excludes 29.2% of lanes in the first match and 21.4% in the second, mostly because of defenders far from the ball (33.5% and 25.6% when adding lanes to offside teammates and crosses, section 6). In the final version, the long-pass rule (section 6) removes a further fifth or so of the remaining lanes (19.5% and 19.9%).

5.5 The 18 matches without pose

The remaining 18 SkillCorner matches, which have tracking and events but no pose, were used only to study the probability of a pass reaching its target, since this does not depend on body orientation. Across the 20 matches there are 14,584 ground passes up to 35 m, of which 85% reach the teammate. The pass outcome given by SkillCorner does not, however, distinguish the reason for failure (it groups interceptions, balls out of play and poorly executed passes), and an approximation (a failed pass after which the next possession starts, on the defending side, less than 5 m from the lane) suggests that only about 11% of failed passes, 239 of 2,196, correspond to a cut near the lane, a finding that bears on how the shadow's value can be interpreted (section 7).

6. Modelling: defining the Cover Shadow Value

The Cover Shadow Value is computed five times per second, at every moment in which an opponent has the ball under control, in steps described below with play C06 as the example (Brisbane Roar–Perth Glory, first half, minute 21, with Perth Glory building up through Pennington).

6.1 Passing lanes and their value

Each moment has one passing lane from the ball carrier to each teammate, allowing only ground passes up to 35 m, since beyond that distance most passes go in the air (among passes with that information, 52% of those between 35 and 40 m and 78% of those over 40 m, against 17% of those between 30 and 35 m). The value of each lane adds two parts. The first is the xT gain between the ball carrier's zone and the receiver's, counting only the positive part, and measures what the pass would give the team in possession. The second is the danger the opponent would gain by winning the ball back at the point where the lane would be cut (the point the defender with the highest cut probability would reach first), read on the same grid from the defending side, and measures what the team in possession would risk by attempting the pass, a part that gives value to closing short passes near one's own goal, where the xT gain is practically zero. Lanes to teammates in an offside position are not evaluated and, when the ball carrier is in a wide channel of the attacking third, lanes into the box only count if they are cut-backs or short passes (up to about 12 m), because the rest are, in practice, aerial crosses. In C06, three lanes longer than 35 m drop out (to Faisal, R. Warland and Cook), and Pennington's remaining seven add up to a value of 0.114, the threat available in the play. This way of valuing each lane (xT gain plus turnover cost) is called, throughout the text, the risk currency, as opposed to the two alternatives evaluated in section 7.6, the xT gain alone and lines broken.

6.2 Who counts

Throughout the text, “defender” means any player of the team out of possession, forwards included. The goalkeeper is not included, and neither is a defender who is marking an opponent, because the voluntary action is the marking and not the shadow. A defender is considered to be marking when he tracks the nearest attacker within 4 m, on the goal side (with a 1 m tolerance, to include side-by-side marking), for at least 1 s, and stops marking only when he moves more than 5 m away. The ball carrier is outside this rule: whoever presses him is not marking and is evaluated by the shadow he casts. In C06, three of the ten defenders are marking (O'Shea, Burke-Gilroy and Waddingham).

6.3 The shadow

Each defender who counts casts a cone behind his body, opposite to the front of the trunk, 2 m wide at the body, with a 60° opening and 10 m long (Figure 2). The length corresponds to the distance behind the defender up to which the ball cannot be played over him, and the initial width represents the reach of the body itself, the 10 m being an estimate, tested in section 7.4. The cone always follows the trunk, even when the defender has his back to the ball carrier (4.1% of observations, with the front of the trunk more than 90° from the direction of the ball carrier), in which case it lies on the ball carrier's side, and these moments account for 8.1% of denied threat, a point taken up in the limitations (section 9.2). A passing lane is only evaluated if all the defenders who can affect it (those within 11 m of it, the cone length plus half the width) have pose.

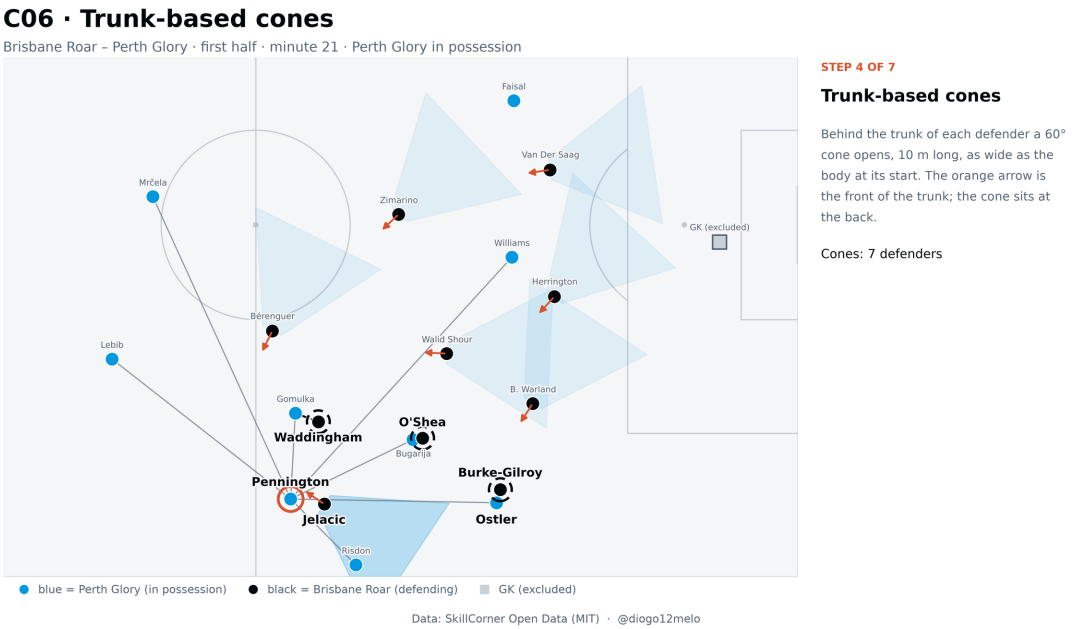


Figure 2. Play C06 (Brisbane Roar–Perth Glory, first half, minute 21): 60°, 10 m cone behind the trunk of each player who is not marking; the arrow shows the front of the trunk.

6.4 Lane arrival index

When a lane crosses a defender's cone, the probability of him reaching some point of that lane before the ball is computed with Spearman's (2018) parameters as implemented by Shaw (n.d.): 0.7 s reaction time, during which the defender keeps the velocity he had, subsequent movement at 5 m/s, ball at 15 m/s and an uncertainty of 0.45 s. Speeds above 12 m/s are treated as tracking errors. To this probability is added the body block: if the lane passes very close to the defender, inside the cone, the ball hits him without him having to react, with a probability that goes from practically 1 at 0.3 m from the body to 0 at 1.2 m, values chosen to reproduce, without an artificial step, the fixed 0.7 m radius used by Cascioli et al. (2025). Being in the cone is therefore not enough: a lane that only crosses the part of the cone far from the defender gets a low probability. Since this probability cannot be calibrated against real passes (section 7), the result is read as an index, and not as the actual probability of the pass arriving.

Several defenders on the same lane are combined as independent: the lane's arrival index is the product, over those defenders, of the probability of each one not reaching the ball first, and it ranges from 1, on an open lane, to 0, on a fully closed lane. In C06, after the long-pass rule, only the lane to Ostler has a shadow, with an arrival index of 0.33 (the lane is two-thirds closed), as shown in Figure 3.

6.5 Denied threat and attribution

The denied threat on a lane is its value multiplied by the drop in the arrival index caused by the shadows, the index without shadows being 1:

$$\textit{denied threat on the lane} = (\textit{xT gain} + \textit{turnover cost}) \times (1 - \textit{arrival index with shadows}) \quad (1)$$

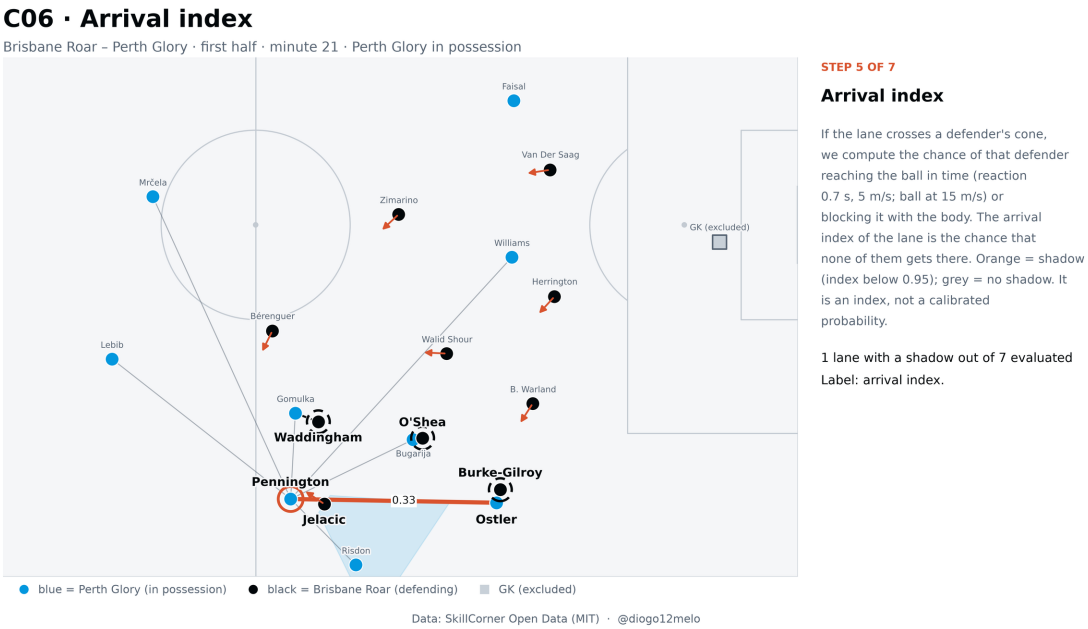


Figure 3. Play C06: arrival index on the only lane with a shadow, to Ostler (0.33: the lane is two-thirds closed).

This expression is the drop in the expected value of attempting the pass. If the pass arrives, the team in possession gains the xT gain and, if it is cut, loses the turnover cost, so the expected value of attempting it is the arrival index times the gain, minus its complement times the cost, a value that, without shadows, equals the gain, the difference between the two scenarios being expression (1). Denied threat therefore measures the potential value that positioning takes away from a pass, if it is attempted, and rests on the assumption that a pass that does not arrive is cut at the cut point (section 5.5), the influence of shadows on the ball carrier's decision being evaluated separately (sections 7.2 and 7.5). Values are on the xT scale (probability of scoring) and are read as an index.

When more than one defender closes the same lane, the value is shared through the Shapley value (Shapley, 1953): each possible order in which the defenders enter is considered, how much each one adds to closing the lane is recorded and the average is taken. With a single defender, the result coincides with comparing the scenario with and without him. In C06 (Figure 4), the denied threat is 0.0109, 9.6% of the threat available in the play, all attributed to Jelacic, whose body and positioning close the outside lane, to Ostler; Zimarino, who in an earlier version added more value through the 44 m lane to Faisal, loses it with the long-pass rule. A player's Cover Shadow Value is thus the denied threat attributed to him, summed over lanes, moments and possessions, and read through the phase index (section 6.6).

6.6 Aggregation

Within each possession the average over moments is taken, so that a lane closed throughout the possession is worth more than one closed for half a second and long possessions do not inflate the total; per player, possessions are summed and the percentage of available threat that was denied is also computed. Totals add up alternative lanes of the same moment, of which only one can be played, so they are read as an index of denied threat and not as goals prevented. The sum was chosen, rather than the highest-value lane, because closing several lanes at once is a greater contribution than closing one, and the alternative that weighted each lane by its probability of being chosen did not meet the criteria set (section 7.8). To compare players, each player's denied threat per moment in a defensive phase (high block, mid block, low block, transition, among others) is computed, counting all the moments in which he defends in that phase, and divided by the per-moment average of all players in the phase, so that 100 corresponds to the average. Whoever closes lanes in a high press is thus compared with those who press high, and not with those who defend near their own box, where lane

values are naturally higher, with only players with at least 100 moments (20 s) in the phase entering the distributions and percentiles (54, 56 and 55 in the high, mid and low block, counting each match separately). The index is therefore an average over the match, not the value of a play, and, since many players close little and a few close a lot, its distribution is skewed (in the mid block, the median is 102 and the top 10% are above 245; in the high block, 61 and 279), so values above 200 are common among the best in each phase and the percentile is the most direct way to read them. Jelacic, for example, defended in a mid block for about 3.7 minutes of opposition possession and, in that time, was ahead of 73% of the 56 players in that phase, at about 1.6 times the average (index of 157).

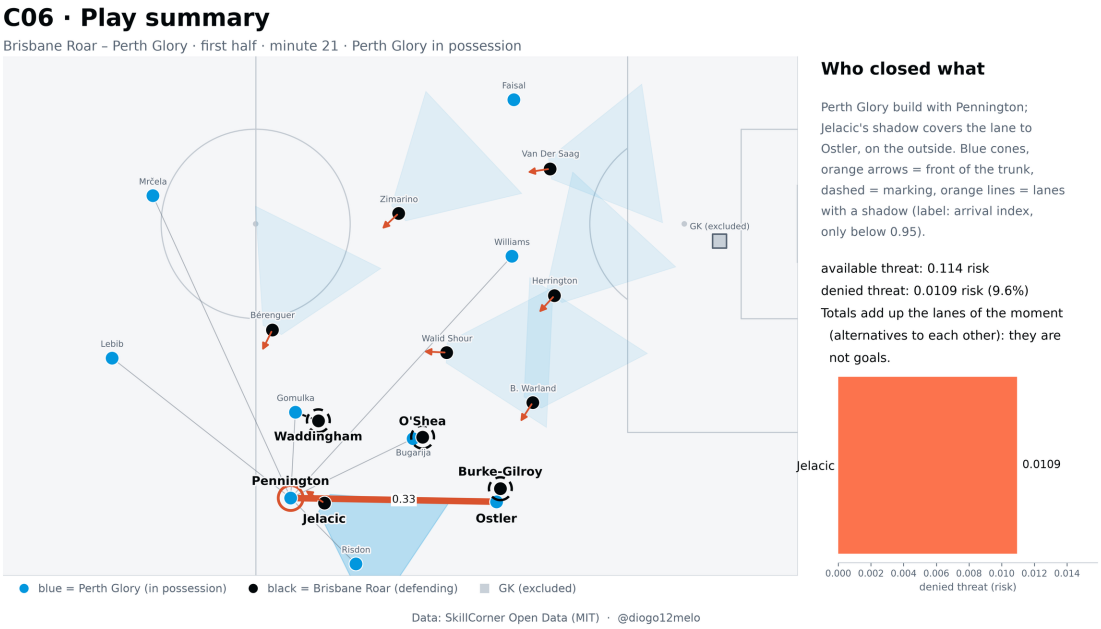


Figure 4. Play C06, summary: denied threat of 0.0109 (9.6% of the available threat), all attributed to Jelacic.

Table 2. Metric parameters and their origin.

Parameter	Value	Origin
Cone opening	60°	Method decision, tested: 45° and 90° do not change the ranking (ρ of 1.000 and 0.996; section 7.4)
Cone length	10 m	Estimate (method decision), tested: 8 m and 12 m do not change the ranking (ρ of 1.000 and 0.996; section 7.4)
Width at the body	2 m	Body reach (method decision), tested in the sensitivity analysis
Reaction time	0.7 s	Spearman (2018), via Shaw
Defender speed	5 m/s	Spearman (2018), via Shaw
Ball speed	15 m/s	Shaw
Time uncertainty	0.45 s	Spearman (2018), via Shaw
Body block	1 at 0.3 m, 0 at 1.2 m	Equivalent, without a step, to the 0.7 m radius of Cascioli et al. (2025); sensitivity in section 7.4
Marking	≤ 4 m, ≥ 1 s, exit > 5 m	Review of plays
Maximum lane length	35 m	Real passes (beyond it, most go in the air)
Lane value	xT gain + turnover cost	Method decision (section 7.6)
Reading	Phase index (100 = average; ≥ 100 moments)	Method decision

7. Evaluation

7.1 Review of plays (K1 and K2)

The metric was reviewed in three rounds, the first by the author, to refine the definition, and the next two by coaches external to the work. In each round, the reviewer saw, for each play, a short clip and the image of the moment analysed, with the shadows drawn, and answered, with written instructions for each answer, two questions: do the shadows make sense, and is the marking correct? Plays were chosen to include those with the highest denied threat, random plays and plays in which the trunk-based cone and the ball-facing cone disagree most (identified with a version of this comparison that had a bug, since fixed, so some of them are not the ones with the greatest disagreement in the final version), without the reviewer, in any of the rounds, knowing which group each belonged to. The results of each round are shown in Table 3 and Figure 5. This is a face-validity check, assessing whether the metric's output is plausible to someone who knows the game, not agreement with an annotation made without seeing the metric.

Table 3. Results of the three rounds of play review (the first by the author, the following ones by coaches external to the work).

Round	Version	Plays	Shadows make sense	Marking correct
1st (author)	v1	15	7 (8 partly)	5 (9 partly, 1 no)
2nd (external coach)	v2	15	15	10 (4 partly, 1 no)
3rd (external coach)	v3	8	8	7 (1 partly)

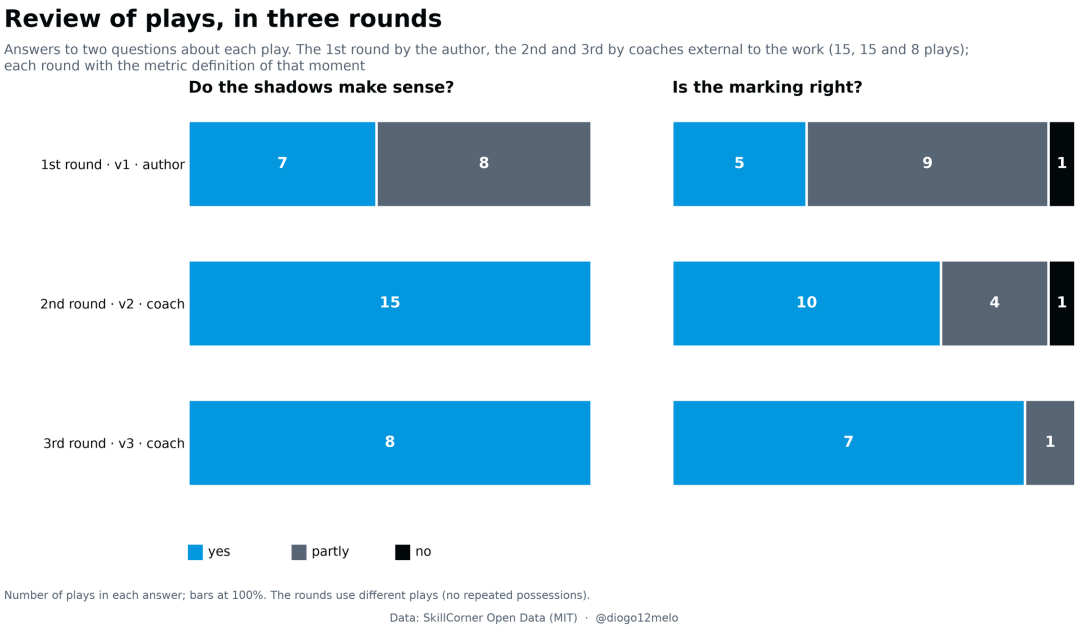


Figure 5. Answers (yes, partly, no) in the three review rounds, the first by the author and the following ones by external coaches, for shadows and marking.

The first round revealed that the five plays with the most denied threat were almost all crosses, situations in which the ball comes in the air and the shadow loses its meaning, and it led to six changes to the definition: excluding the goalkeeper and lanes to offside players, handling crossing zones, separating pressing from marking, continuity of marking and the body block. The second round led to widening the marking distance and to the gradual body block. The third, carried out with new plays after all adjustments, works as an independent validation, and the only partial case (a defender 0.45 m outside the goal-side tolerance) was considered by the reviewer himself a matter of interpretation.

Before the remaining criteria, it is worth seeing how denied threat is distributed across the defensive phases, since lane values depend heavily on the area of the pitch (Figure 6). Within each phase, the share of available threat that is denied is 12.9% in the high block and 14.0% in the mid block, the phases of pressing the opponent's build-up, and 30.7% in the low block. Of the total denied threat in the two matches, the high and mid blocks together account for 42.2% (22.0% and 20.2%), against 34.8% for the low block.

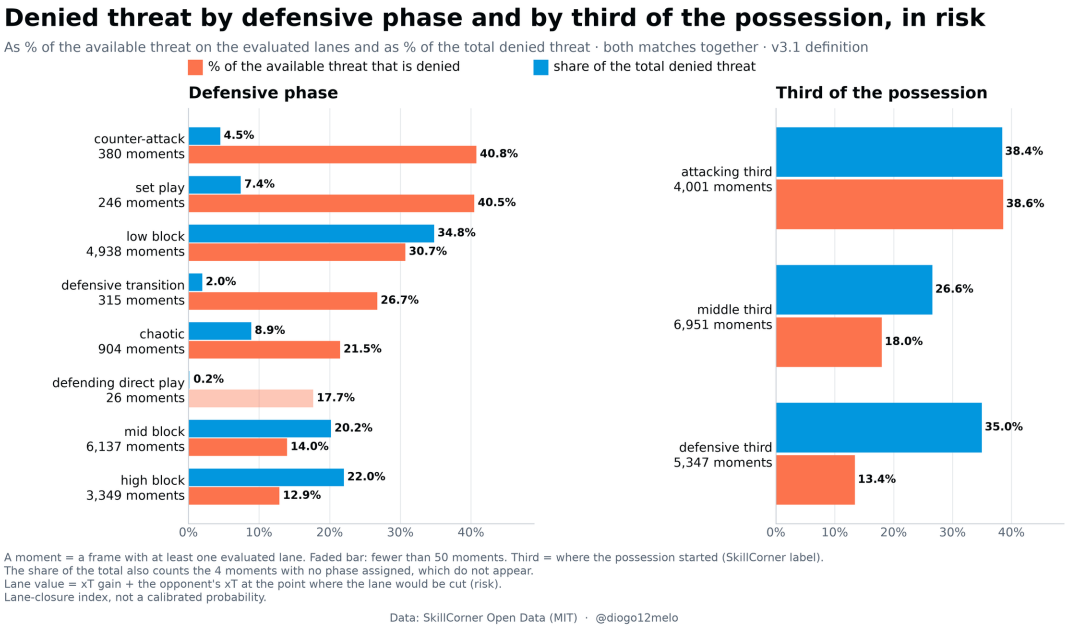


Figure 6. Denied threat as a percentage of available threat, by defensive phase and by third where the possession started (two matches, final version, risk currency).

7.2 Coherence with real passes (K3)

If the shadow measures what it claims to measure, players should avoid shadowed lanes. Comparing each pass made with the alternatives of the same moment with similar length and danger, the chosen lane has less shadow than the alternatives, with a difference of −0.118 (95% CI [−0.137; −0.099], 1,071 passes, one per possession, bootstrap interval; Figure 7); without that control, the difference is −0.117.

The metric does, however, overstate how closed lanes are: of the passes attempted on lanes with an arrival index below 0.5, 63.1% arrived (53 of 84). Part of this gap is explained by only observing the passes the player decided to risk, since a passer only attempts a shadowed lane when he sees the ball will get through, so this check is not enough to calibrate the index as a probability, a question taken up again in section 7.7.

7.3 The contribution of the body (K5)

Compared with the same metric using a ball-facing cone (assuming each player looks at the ball carrier, with the cone behind him on the side opposite the ball), trunk orientation changes who casts a shadow in 36.6% of evaluated lanes and slightly reduces denied threat (19.5% against 21.0% in the first match and 19.8% against 20.9% in the second). The ranking of players remains similar (Spearman's ρ of 0.942; Figure 8, or 0.928 when comparing denied threat per moment, which removes the weight of minutes played), the minimum across positions (0.400) being for right centre-backs, with only four cases (player per match), where a single swap in the ranking is enough for ρ to drop to that value. In C06 (Figure 9), the ball-facing cone triples the denied threat (from 0.0109 to 0.0327) and shares it differently, with Walid Shour becoming the main contributor (0.0190) and Jelacic adding 0.0138.

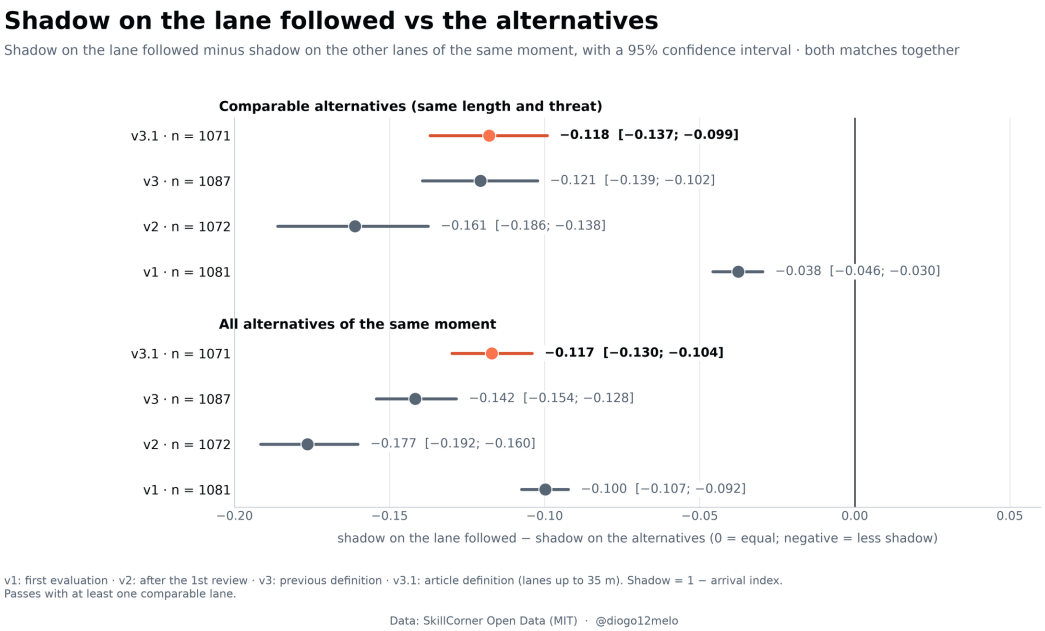


Figure 7. Difference between the shadow on the chosen lane and on the alternatives of the same moment, with 95% confidence interval (bootstrap), by metric version (v1, v2, v3 and v3.1).

The comparison depends on where players are actually facing. In 95.9% of observations, the front of the trunk is within 90° of the direction of the ball carrier and only in 0.6% does it exceed 135° (n = 124,671, excluding goalkeepers, five observations per second), full-backs being the ones who face the ball least (49.9% of observations within 45°, against 71.0% for forwards). Since a player facing the opponent's half is often also facing the ball, the instants in which the two directions diverge were also compared (Figure 16), and when they differ by more than 90° the trunk is closer to the direction of the ball carrier in 96.7% of cases (median of 31° relative to the ball carrier and 143° relative to the forward direction; n = 32,731).

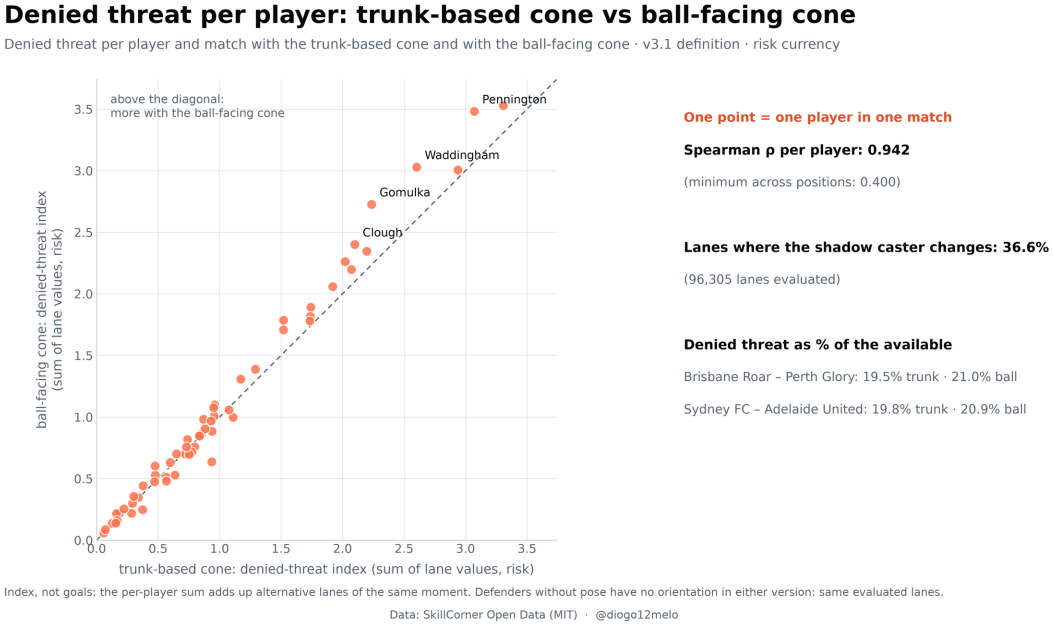


Figure 8. Denied threat by player and match with the trunk-based cone and with the ball-facing cone (final version, risk currency).

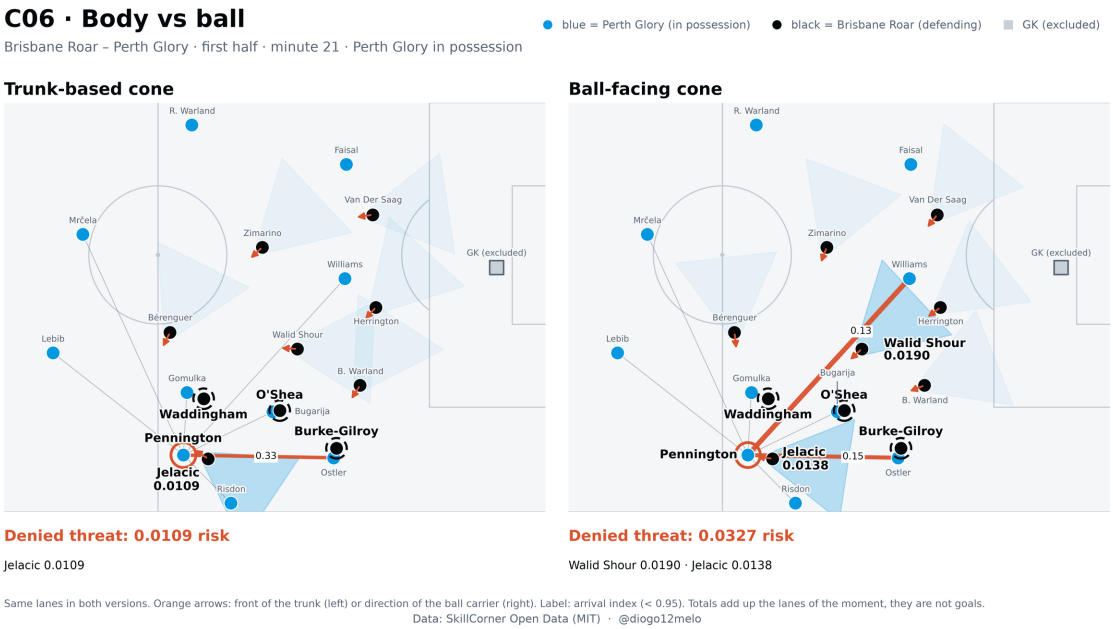


Figure 9. Play C06 with the trunk-based cone (the metric) and with the ball-facing cone: 0.0109 against 0.0327 of denied threat.

7.4 Stability (K4)

With reasonable variations of the cone's opening and length, the width at the body, the times, the speeds and the marking thresholds, the ranking of players within each position barely changes (mean ρ between 0.955 and 1.0; Figure 10), and the same holds for the lane length limit (0.971 at 30 m and 0.996 at 40 m) and for the way of aggregating over time, since, considering only the last second, or the last two, before the end of each individual possession, ρ is 0.942 and 0.960. The body block is the exception in absolute value, which ranges from 7.5% without a block to 27.7% with a fixed 1.5 m radius (first match), but not in the ranking, except when it is removed altogether (ρ between 0.956 and 1.0 with radii from 0.5 to 1.5 m, against 0.875 without a block). In these variants, the turnover cost stays as in the base version, only the shadows changing. These ρ values compare each player's total denied threat, which also depends on minutes played, and, with denied threat per moment, which removes that weight, the ranking remains stable across the parameter and lane-length variants (ρ between 0.928 and 1.0), except when the body block is removed altogether, in which case ρ drops to 0.31. The metric should therefore be read comparatively.

7.5 The shadow and pass choice (K8)

If the shadow conditions the ball carrier, it should help predict who the pass goes to. To test this, a discrete choice model (conditional logit) was fitted in which, for each pass, the ball carrier chooses one of the available lanes based on length, progression towards goal, xT gain, whether the receiver is marked and the receiver's distance to the nearest opponent, with and without each lane's shadow. With 1,414 ground passes up to 35 m, fitting the model on one match and testing it on the other, the trunk-based shadow improves prediction in both directions (log-loss per pass from 1.478 to 1.419 and from 1.534 to 1.480; Figure 11), and a fully shadowed lane has about one sixth of the probability of being chosen of an open lane with the same characteristics (odds ratio of 0.17, in the model with the trunk-based shadow only). When both shadows enter the same model, the trunk-based one keeps its weight (−1.36, 95% CI [−1.94; −0.78]) and the ball-facing one can no longer be distinguished from zero (−0.41, 95% CI [−0.93; 0.10]), indicating that body orientation holds information about the ball carrier's decision that the position relative to the ball does not. The out-of-sample advantage over the ball-facing shadow is, however, small and, in one direction, cannot be distinguished from zero (95% bootstrap interval over passes). No causal relationship follows from this, not least

because the shadow is measured at the moment of choice, within 0.1 s of the pass, and defenders also position themselves in anticipation of the pass. The choice of the trunk as the origin of the shadow stems, however, from the reading of the game and not from this test, since it is the defender's body that closes the lane. K8 serves to check whether the measurement behaves as that reading predicts, which, in the two matches analysed, holds for the ball carrier's decision, albeit without a conclusive advantage over the ball-facing shadow.

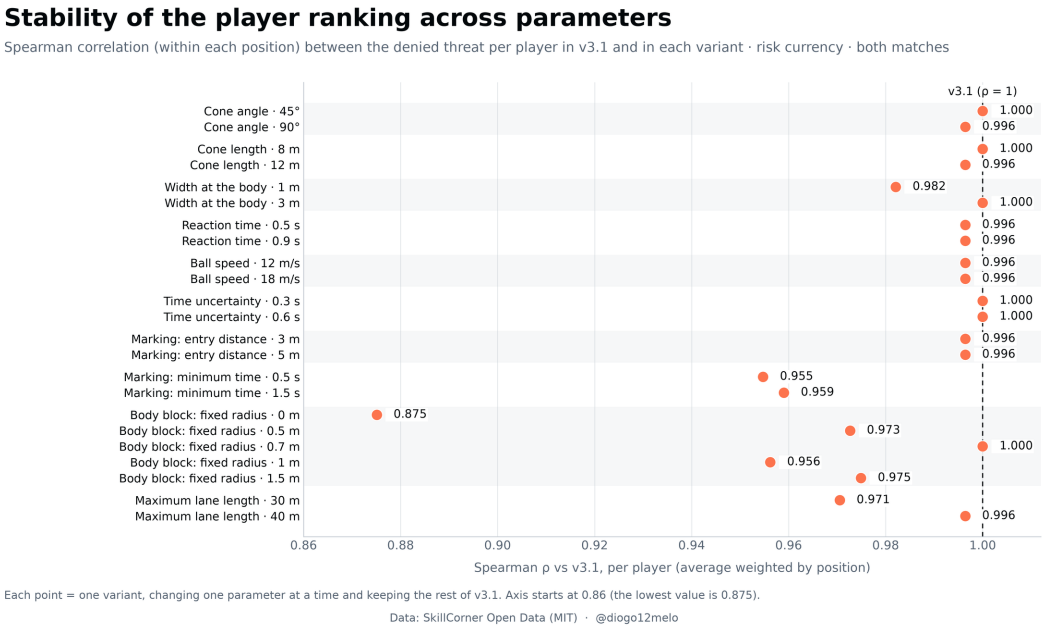


Figure 10. Spearman's ρ of denied threat per player, within position, between the final version and 23 parameter variants (section 7.4).

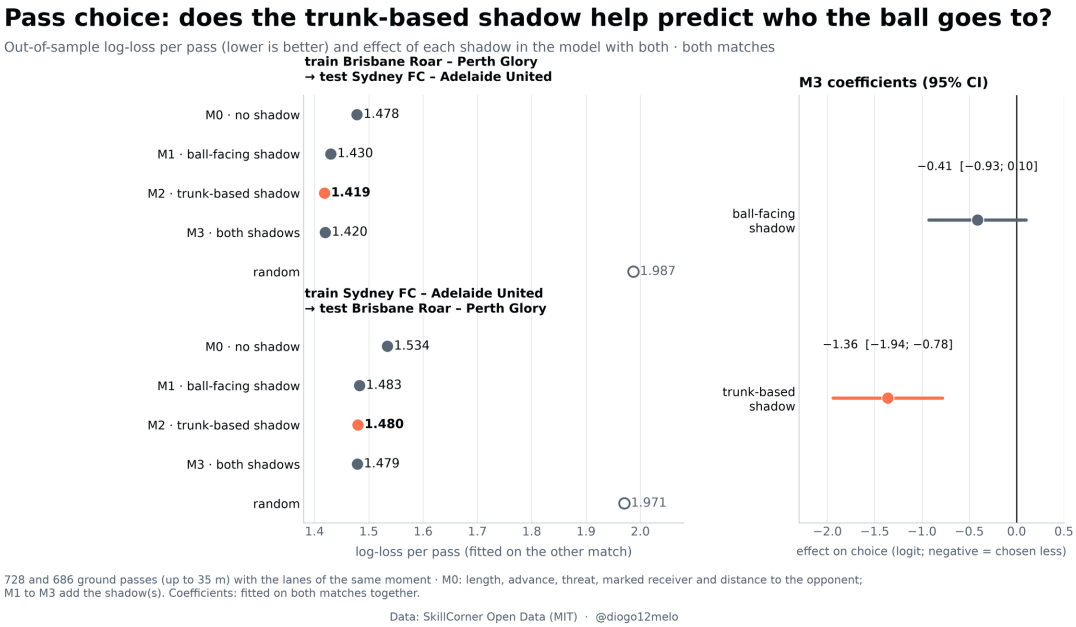


Figure 11. Out-of-sample log-loss per pass of the pass choice models (no shadow, ball-facing shadow, trunk-based shadow and both), in both directions, and coefficients of the two shadows in the joint model, with 95% confidence intervals.

7.6 The value of each lane: three currencies

The value of a lane was measured in three ways: the xT gain, the number of opponents the pass would bypass (lines broken) and the sum of the xT gain and the danger the opponent would gain by winning the ball

back at the cut point (risk). Within each defensive phase, the three rank players in a similar way (per-player ρ between risk and xT of 0.82 in the high block, 0.95 in the mid block and 0.90 in the low block), but they distribute value very differently across phases (Figure 12), with the high and mid blocks accounting for 11.3% of denied threat with xT, 54.6% with lines broken and 42.2% with risk.

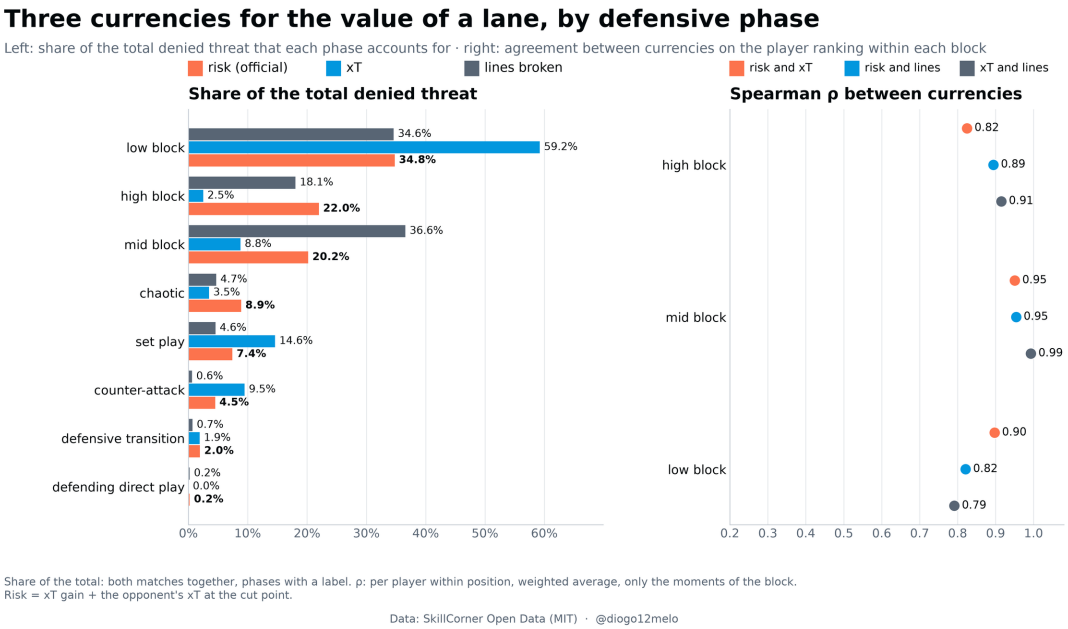


Figure 12. Share of total denied threat accounted for by each defensive phase under each of the three currencies (risk, xT and lines broken) and per-player Spearman's ρ between currencies, within each block.

The difference shows in the plays. When a forward blocks the goalkeeper's lane, inside his own box, to a centre-back 6 m away (Figure 13), xT gives him zero, because the pass hardly changes the ball's zone, and lines broken little more, while risk gives him 73% of the threat available in the play, since a cut at that point would give the opponent a chance right in front of goal; conversely, a cut-back to the centre of the box bypasses no opponents and is worth zero in lines broken, even though it is among the most dangerous. Risk was adopted, being the only one of the three that values both situations, with two caveats: the turnover cost comes from a 12 × 8 grid, which treats any cut in front of goal in the same way, and the value is potential, that is, it measures what the pass would cost if it were attempted.

7.7 The shadow and pass success

It remained to be seen whether the shadow also makes the pass less likely to arrive, as the index in section 6.4 assumes. A first attempt to calibrate the physical interception model against real passes, first with the two matches and then with all 20, failed the criteria fixed before computation: the fitted parameters ended up at the limits of the search (a reaction time of 0.05 s and a speed of 2 m/s, with no physical meaning) and the calibration error (ECE, with ten bins by decile of predicted probability) was 0.066, against a maximum of 0.03, which is consistent with most failed passes not resulting from a cut (section 5.5). A simpler statistical model (xPass), with geometric variables and no body orientation, calibrates very well across the 20 matches (calibration error of 0.005 and a log-loss better than always predicting the base rate, in every match; Figure 14). Adding the shadow to it in the two matches with pose, its weight is negative, as expected, but the interval includes zero (−0.37, 95% CI [−0.95; 0.21]) and out-of-sample prediction does not improve. One possible conclusion is that, once the defenders' proximity to the lane is known, body orientation weighs more on the ball carrier's decision (section 7.5) than on the probability of the pass arriving, a conclusion limited by there being only about 200 failed passes in the two matches with pose.

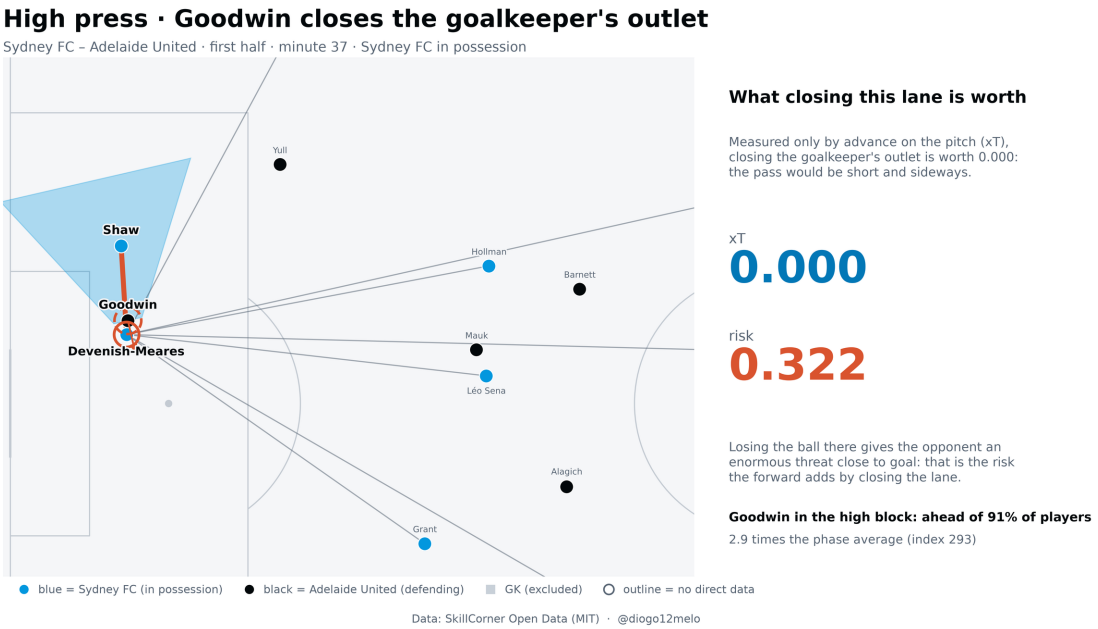


Figure 13. High press (Sydney FC–Adelaide United, first half, minute 37): Goodwin, of Adelaide United, closes the lane from the goalkeeper to the centre-back, worth 0.000 in xT and 0.322 in risk.

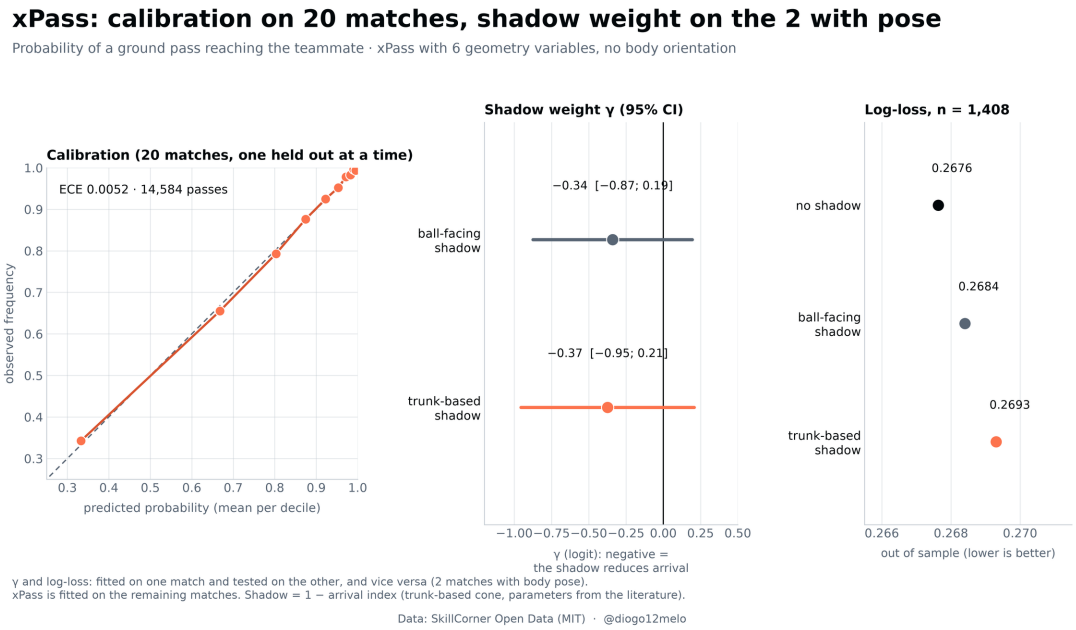


Figure 14. xPass calibration with 20 matches (leave-one-match-out), weight γ of the shadow in the two matches with pose, with 95% confidence interval, and out-of-sample log-loss with and without shadow.

7.8 A version with interception and deterrence

A more complete version was also tested, along the lines of Kim et al. (2025), in which each lane was weighted by its probability of being chosen and each player's value was split between cutting the lane (interception) and making the ball carrier change his mind (deterrence). With criteria fixed before computation, the version was not adopted: in between 31% and 71% of moments, depending on the currency, the shadow appeared to increase danger, because the model forces the ball carrier to always pass and does not consider the possibility of keeping or losing the ball, and the calibration of pass success with the shadow fell slightly short of what was required. With only two matches with pose, this route is left as future work.

7.9 Reproducibility and explainability (K6 and K7)

The final version exactly reproduces the tables computed during modelling, and the full pipeline, run from scratch in a clean environment from the original data, gives the article's numbers again, with both matches computed with the same parameters. Explainability (K7), defined in section 4.1 as a coach's ability to understand the animation of a play in under a minute, was evaluated with three coaches who did not take part in the review of plays, who were shown the C06 animation without prior explanation. All three understood the purpose of the metric in under a minute (the only one who had seen an earlier version found the final one more explicit), although the detailed workings and the reading of the values only became clear after an explanation, so the animation is enough to convey the idea but does not replace guidance when it comes to interpreting the numbers.

7.10 Evaluation against the success criteria

Table 4. Evaluation of the metric against the success criteria.

Criterion	Verdict	Justification
K1 · face validity	Met	8 of 8 plays in the independent review
K2 · separates shadow from marking	Met	7 of 8; the remaining case is a matter of interpretation
K3 · coherence with the game	Partly met	Distinguishes the lanes avoided (−0.118), but the index is not calibrated as a probability
K4 · stability	Met, with one caveat	Stable ranking; the absolute value depends on the body block
K5 · the body changes the shadow	Met	Changes who casts the shadow in 36.6% of lanes and reduces denied threat compared with the ball-facing cone
K6 · reproducible	Met	Results reproduced from scratch; same parameters in both matches
K7 · explainable	Met	Three coaches understood the purpose without prior explanation; reading the values benefits from guidance
K8 · the body influences the decision	Partly met	Predicts better than without shadow; against the ball-facing shadow, the joint model attributes the information to the trunk, but the out-of-sample advantage is significant in only one direction

8. Tactical commentary

The winger example given in the introduction, who can stay wide or position himself inside, has a direct counterpart in play C06, albeit with the opposite option. With the ball at Pennington, Jelacic (Brisbane Roar's left winger) positions himself and orients his body so as to close the outside passing lane, to Ostler, and to steer Perth Glory's build-up inside, an area where, on a reading of the play, the team would be prepared to press and win the ball, a kind of inside trap. All the denied threat in the play comes from that lane (0.0109) and, over the match, in the mid block, Jelacic is ahead of 73% of players (index of 157, section 6.6). Zimarino, the right winger, also closed the lane to Faisal, but a 44 m ground pass would hardly ever happen, a coach's reading that led to the long-pass rule and leaves Jelacic as the only player with value in the play. Closing inside or outside depends, however, on the team's game plan, there being situations in which the opponent is steered wide, as in the introduction's example, and others, like this one, in which he is steered inside. It does not follow that the metric assesses whether that choice fits the plan, not least because the value of each lane is given by the danger it carries (the xT gain and the turnover cost) and not by the team's intention, so the metric only measures what each player closes within the option taken.

Play C03 (Sydney FC–Adelaide United, second half, minute 19) shows the same action in the first line of pressure (Figure 15). With the ball at Vriends, Adelaide's centre-back, the passing lane to Isaías, the defensive midfielder, lies inside the cone behind the trunk of Klimala, Sydney's forward, with an arrival index of

0.12, which gives Klimala 13.4% of the threat available in the play. Over the match, in the mid block, Klimala is ahead of 86% of players, at about 2.2 times the phase average (index of 218). The reading of the image is that, from this position, the forward can press the centre-back and at the same time take away his lane to the defensive midfielder, that is, one player conditioning two. From there he can hold his position, closing the midfielder's lane and leaving the centre-back to decide to one side, the other or long, or go into a low-intensity press that tries to force that decision, and in either case Klimala's position and orientation are compatible with the intention to close the defensive midfielder's lane. This is the kind of contribution that, as noted in the introduction, traditional defensive metrics do not record, since neither option involves, in itself, a tackle or an interception.

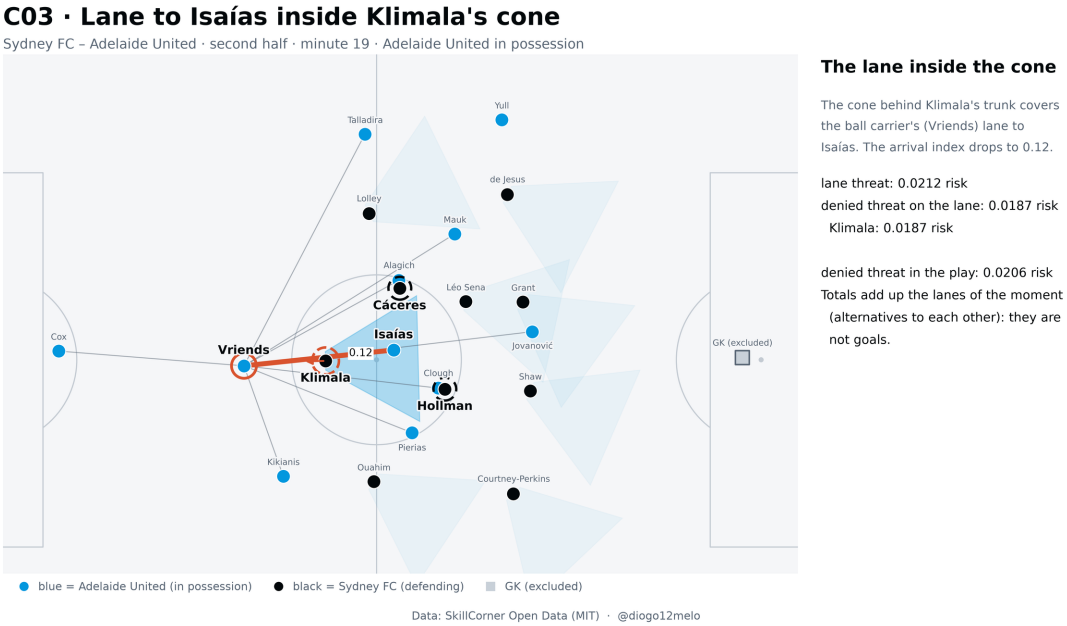


Figure 15. Play C03 (Sydney FC–Adelaide United, second half, minute 19): the lane from Vriends to Isaías inside Klimala's cone, with an arrival index of 0.12.

The comparison with the ball-facing cone, presented in section 7.3, showed that, in C06, that alternative triples the denied threat and moves most of the value to Walid Shour, whose trunk is not oriented to close the lane. One possible reading is that players of the team out of possession are normally oriented towards the ball side, so that those whose job is not to close a lane do not close it with their body, a difference a ball-facing cone does not capture. The orientations measured in the two matches (section 7.3 and Figure 16) are consistent with the first part of this reading. The data, however, only describe orientation, so the link between each player's orientation and his role at that moment remains a reading, compatible with what is observed but yet to be demonstrated. The results of sections 7.5 and 7.7 nonetheless add something to it: the trunk-based shadow helps predict where the ball carrier passes, while the data do not allow a conclusion on whether it also weighs on pass success, which is compatible with the defender's body acting above all on the ball carrier's decision.

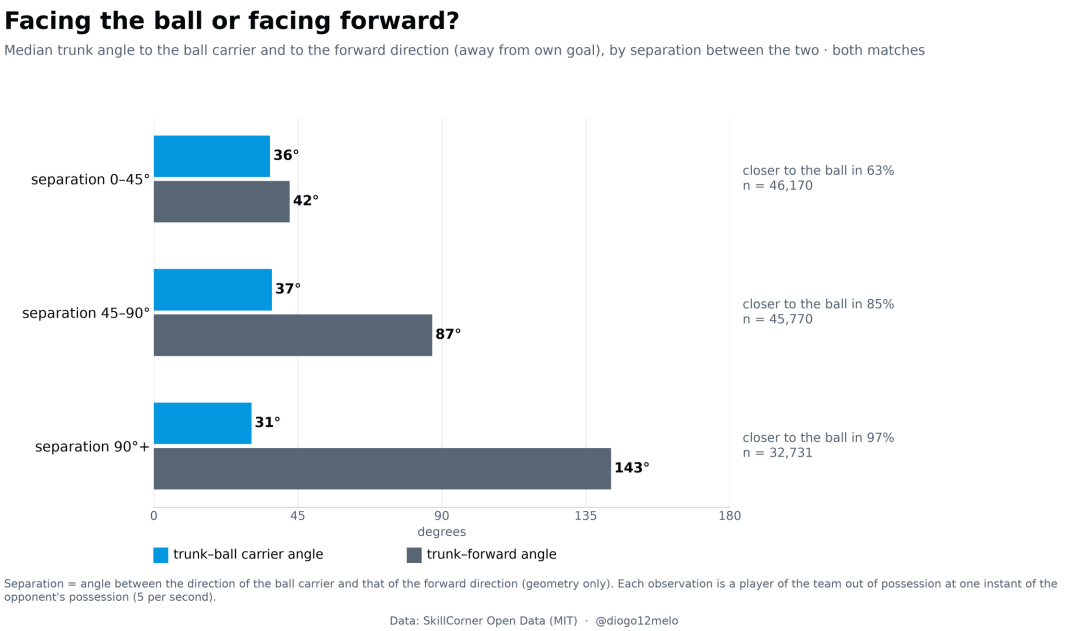


Figure 16. Median angle of the trunk relative to the ball carrier and to the forward direction, by separation between the two directions (players of the team out of possession with pose, excluding goalkeepers, five observations per second).

The Goodwin case (Figure 13) shows the same action at the start of the build-up. With the ball at the goalkeeper, inside his own box, the forward positions himself so as to block with his body the short exit to the centre-back, closing one side of the build-up and steering it towards where, on a reading of the play, the team would be prepared to press. In the adopted currency, this action is worth 73% of the threat available in the play, a value that the xT gain alone does not see (section 7.6). Over the match, in the high block, Goodwin is ahead of 91% of players, at about 2.9 times the phase average (index of 293).

For a coach or analyst, the metric makes it possible to review, play by play, which lanes each player closed and what they were worth, which can support video feedback or the comparison of players in the same defensive phase, through the phase index (section 6.6), which allows, for example, comparing only those who press high. Absolute values should, however, be treated with caution, since, as seen in sections 7.2 and 7.7, the arrival index behind them is not calibrated.

9. Conclusions, limitations and future work

9.1 Conclusions

The results from the two matches analysed indicate that it is possible to value, with tracking and pose data, a defensive action that until now was perceptible mainly through observation: conditioning the opponent through off-ball positioning and body orientation. On the first question, the shadow is defined as a cone behind the trunk of each defender who is not marking, which closes a lane to the extent that the defender would reach it before the ball or block it with his body, and, in the last review round, an external coach considered the shadows coherent in 8 of 8 plays and the marking correct in 7 of 8, in a face-validity check. On the second, each lane's value adds the cost of losing the ball at the cut point to the xT gain, the drop in the arrival index measures what the shadows take away from the pass and the Shapley value shares it among the defenders who close the same lane, also giving value to the work of those who press the opponent's build-up, which the xT gain alone hardly sees. In the passes actually made, the chosen lane has less shadow than comparable alternatives, which is coherent with players' choices.

On the third question, compared with a ball-facing cone, the trunk changes who casts a shadow in 36.6% of lanes, without materially changing the ranking of players (ρ of 0.942). On the fourth, that ranking remains

stable under reasonable variations of the parameters, also when comparing denied threat per moment, except when the body block is removed altogether. On the fifth, the trunk-based shadow helps predict where the ball carrier passes and, in a model with both shadows, keeps the weight the ball-facing shadow loses, although its out-of-sample advantage over the latter is small, and the data do not allow a conclusion on whether it also weighs on pass success, which is compatible with the defender's body acting above all on the ball carrier's decision.

9.2 Limitations

The heaviest limitation is the data. In the open data, pose exists for only two matches of one league, and only for the players the broadcast camera captures (64.5% and 73.1% of cases), so lanes in which some relevant player has no pose are left out (29.2% and 21.4%), mainly those that depend on players far from the ball. The results therefore describe mostly what happens near the ball and cannot be generalised to other matches, teams or leagues.

The second limitation concerns calibration. The arrival index cannot be read as the probability of the pass arriving, since, of the passes attempted on lanes with an index below 0.5, 63.1% arrived, and two attempts to calibrate it against real passes, with two and with 20 matches, failed, partly because most failed passes do not result from a cut. The absolute value also depends on the body block (from 7.5% to 27.7% of available threat in the first match, section 7.4), a parameter that, like the times and speeds, comes from the literature or from method decisions. Values should therefore be read comparatively, through the phase index, and not as goals prevented.

There are also limits in the definition itself and in the validation. The metric only considers ground passes up to 35 m and lanes to each teammate's current position, leaving out passes into space, the turnover cost comes from a 12 × 8 grid, the value is potential (what it would cost if the pass were attempted) and SkillCorner's defensive phases are assigned to a whole phase rather than to each moment. Defenders on the same lane are combined as independent, even though they contest the same ball, the error estimate for each pose joint does not enter the computation and the cone follows the trunk even when the defender has his back to the ball carrier, in which case it lies on the ball carrier's side (8.1% of denied threat, section 6.3), a situation that, in pressing, does not match the tactical idea of a shadow. The body enters only through the cone, since the model's reaction time is the same whether the defender is facing the lane or has his back to it, and, since pose is missing above all for players far from the ball, whose moments without evaluable lanes count as zero, the phase index inherits this camera limitation. The value of each lane knows nothing about the team's plan, so it does not assess whether the lane closed was the one the plan called for (section 8), and trunk orientation describes the body, not the player's intention. The review of plays assessed the face validity of the shadows, that is, whether the metric's output seemed plausible to those who saw it, and not agreement with an annotation made without seeing it, and, by coaches external to the work, was limited to two rounds, with explainability assessed informally, with three coaches.

9.3 Future work and applications

Since the aim was to demonstrate a method, not to evaluate players, validation with more data is left to those who have it. With body pose for full seasons, it would be possible to return to the version that separates interception and deterrence (section 7.8), along the lines of Kim et al. (2025) but with body orientation, adding to the choice model the possibility that the ball carrier does not pass, to estimate more precisely the effect of the shadow on pass success and to check whether the metric remains stable across matches and leagues. More immediate is extending the review of plays to more coaches, including an annotation of closed lanes made without seeing the metric, and making the animation more self-explanatory in how the values are read,

since, in the explainability assessment, the purpose of the metric was understood without help, but the numbers needed explaining.

For those working in football, the interest of a metric of this kind lies in putting a number on a contribution that today depends almost entirely on observation. An analyst can use it to review, play by play, which lanes each player closed and compare that record with what the game plan asked for, and a coach can base feedback on off-ball positioning and orientation on it. With enough data, scouting could also identify players whose defensive value does not show up in tackles or interceptions, like the winger mentioned in the introduction, with few recorded defensive actions and yet essential to the way the team defends.

Declarations

Data availability. The data used are the SkillCorner Open Data (SkillCorner, 2025), public and under the MIT licence.

Code availability. The code that computes the metric, the evaluations and the figures is available at github.com/MeloDiogo12/cover-shadow-value under the MIT licence, together with a description of how to reproduce the results from the original data.

Acknowledgements. To the coaches who reviewed the plays and assessed the animation.

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Use of artificial intelligence tools. An artificial intelligence tool (Claude, by Anthropic) was used to support code writing and text revision. The definition of the metric, the method decisions and the tactical reading are the author's, who reviewed all the content and takes responsibility for it.

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